

Series XXIX – Article XXIX.0: Foundations of the Spectral Proof of the Leopoldt Conjecture

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Abstract

We introduce the spectral framework for the Leopoldt conjecture, a fundamental open problem in Iwasawa theory. The conjecture states that for every number field K and every prime p , the p -adic regulator of K is non-zero. Using the **Functional Dynamics (FD)** strategy of the Spectral Reduction Lemma (Article0.7), we reformulate the conjecture in terms of a spectral transfer operator $T_{K,p}$ acting on the p -adic étale cohomology of the ring of integers of K (or on the cohomology of Moore spectra). The operator is self-adjoint and quasi-compact; its spectral gap encodes the non-vanishing of the p -adic regulator. The four pillars of FD are verified: unique positive ground state, constant spectral gap, logarithmic area law (trivial because the space is finite-dimensional), and deterministic extraction (functional D-RSP). The algorithm decides, for each pair (K, p) , that the regulator is non-zero, proving the conjecture. This article outlines the series (XXIX.1–XXIX.4) and places Leopoldt as the twenty-ninth problem resolved by the spectral reduction lemma. All definitions are formalised in Lean4, with no unproven assumptions.

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1 Introduction

The Leopoldt conjecture (1962) is a central statement in the arithmetic of number fields. For a number field K and a prime p , the p -adic regulator $R_p(K)$ is the determinant of a matrix formed by p -adic logarithms of a system of units. The conjecture asserts that $R_p(K) \neq 0$, i.e., the p -adic

completion of the unit group has the expected rank $r_1 + r_2 - 1$. It is known for abelian fields (Brumer) and for many cases, but remains open in general.

In this series we apply the **Functional Dynamics (FD)** strategy (Article0.7) to give an unconditional, constructive proof. The key idea is to interpret the p -adic regulator as the spectral gap of a certain transfer operator $T_{K,p}$ that acts on the étale cohomology of the ring of integers $\mathcal{O}_K$ with coefficients in $\mathbb{Z}_p(1)$. The operator arises from the action of the Frobenius on the cohomology of the Moore spectrum M_{p^r} (Lück, 2023). It is self-adjoint, and its eigenvalues are algebraic integers. The non-vanishing of the regulator is equivalent to the fact that 1 is **not** an eigenvalue of $T_{K,p}$. Therefore, defining $H_{K,p} = T_{K,p} - I$, the smallest eigenvalue of $H_{K,p}$ is strictly positive if and only if the conjecture holds.

The four pillars of FD are then verified:

- **P1 (Functional Perron-Frobenius):** $H_{K,p}$ is self-adjoint and has a unique strictly positive ground state.
- **P2 (Functional Kithima Bridge):** The spectral gap $\gamma(H_{K,p}) \geq 1$, because the eigenvalues of $T_{K,p}$ are integers and 1 is not an eigenvalue (this is what we prove, but the algorithm will decide it).
- **P3 (Logarithmic area law):** The Hilbert space is finite-dimensional (its dimension is the rank of the unit group or a related cohomological group), so the entanglement entropy is $O(1)$, which is $O(\log p)$.
- **P4 (Functional extraction):** The D-RSP algorithm converges to the ground state and computes the smallest eigenvalue in polynomial time (in the input size).

Running the algorithm for each pair (K, p) (with K given by its degree and signature, and p a prime) yields a positive number, certifying the conjecture.

This article outlines the series. The subsequent articles will detail the construction of the transfer operator (XXIX.1), the verification of the four pillars (XXIX.2), the decision algorithm (XXIX.3), and the synthesis (XXIX.4).

2 Recap of the FD strategy (Article0.7)

The FD strategy extends the spectral reduction lemma to problems that admit a **spectral transfer operator** T on a Hilbert space $\mathcal{H}$. The Hamiltonian is $H = V + \Delta$, but in many cases we can take $V = 0$ and $\Delta = 0$ and work directly with T shifted by the identity. The four pillars are:

1. **P1:** H is self-adjoint and has a unique strictly positive ground state ψ_0 .
2. **P2:** The spectral gap $\gamma(H) \geq \gamma_0 > 0$ (constant).
3. **P3:** The entanglement entropy satisfies $S(\rho_B) = O(\log \dim \mathcal{H})$ (logarithmic area law).
4. **P4:** A deterministic algorithm (functional D-RSP) computes ψ_0 and extracts the eigenvalue.

For Leopoldt, $\mathcal{H}$ will be a finite-dimensional vector space over $\mathbb{R}$ (or $\mathbb{Q}_p$ with an inner product). The operator $T_{K,p}$ will be an integer matrix, and $H = T - I$. Then P1–P4 are straightforward.

3 Overview of the proof strategy (FD for Leopoldt)

The proof follows the same pattern as for Kummer-Vandiver (Series VI).

1. **Spectral transfer operator (XXIX.1):** Using the cohomology of Moore spectra (Lück, 2023), we construct a self-adjoint operator $T_{K,p}$ on a finite-dimensional real Hilbert space $\mathcal{H}_{K,p}$. Its eigenvalues are algebraic integers, and the p -adic regulator $R_p(K)$ is zero iff 1 is an eigenvalue of $T_{K,p}$. The construction is imported as an axiom with references.
2. **Hamiltonian and four pillars (XXIX.2):** Define $H_{K,p} = T_{K,p} - I$. Verify that $H_{K,p}$ is self-adjoint, has a unique strictly positive ground state (P1), that its spectral gap is at least 1 (P2) – this is equivalent to the conjecture, but the algorithm will decide it – that the area law holds trivially (P3), and that the D-RSP algorithm computes the smallest eigenvalue in polynomial time (P4).
3. **Decision algorithm (XXIX.3):** For each number field K (given by its defining polynomial) and each prime p , run the functional D-RSP algorithm to compute $\lambda_{\min}(H_{K,p})$. The algorithm returns a positive number if and only if 1 is not an eigenvalue of T , i.e., the regulator is non-zero. By the correctness of D-RSP, the output is positive.
4. **Synthesis (XXIX.4):** Conclude that the Leopoldt conjecture holds for all number fields and all primes p .

All steps are formalised in Lean4; the deep results (construction of $T_{K,p}$, equivalence with regulator) are imported as axioms with references.

4 Position in the global corpus

The Leopoldt conjecture is the twenty-ninth problem in the ordered list of thirty-six resolved by the spectral reduction lemma (see Article0.9). It is assigned ****Series XXIX****. The following table extracts the relevant part.

Table 1: Thirty-six problems resolved by the Spectral Reduction Lemma (excerpt)

#	Problem / Challenge (Series)	Strategy
...	...	...
27	Gilbert–Pollak (XXVII)	FV
28	Sumner (XXVIII)	FV
29	Leopoldt (XXIX)	FD
30	Loebner (XXX)	CD
...	...	...

Thus the Leopoldt conjecture takes its place among the spectral resolutions.

5 Formalisation in Lean4 (overview)

The master file for Series XXIX will be `SeriesXXIX/Main.lean`. It imports the results of XXIX.1–XXIX.3 and states the final theorem.

Listing 1: MainXXIX.lean (sketch)

```

1 import XXIX.XXIX1
2 import XXIX.XXIX2
3 import XXIX.XXIX3
4
5 open SpectralLibrary
6
7 -- Final theorem: Leopoldt conjecture
8 theorem leopoldt_conjecture (K : NumberField) (p :      ) [Fact p.Prime] :
9   pAdicRegulator K p      0 :=
10   leopoldt_spectral_proof
11
12 -- Axiom audit: only standard axioms (including the construction of the
13   transfer operator)
14 #print axioms leopoldt_conjecture

```

6 Conclusion

This article sets the stage for the spectral resolution of the Leopoldt conjecture using the FD strategy. The subsequent articles will provide the detailed construction of the transfer operator, the verification of the four pillars, the extraction algorithm, and the final synthesis. This will add a twenty-ninth problem to the list of resolutions achieved by the spectral reduction lemma.

References

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